



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTPP/OPERATION/2023-24/LMI REV/471660

Date:11/09/2023(06:22:46)

Subject:REVISION 9 of LMI/OD/OPS/SYST/005

Respected Sir,

This LMI on "Procedure to be followed for Regulating/ Bypassing Permissive And Protections." is revised inline with Corporate Operation Directive "COS-ISO-00-OD/OPS/SYST/005 Rev. No.: 3 Date: AUGUST 2023"

Revision details:

The following change is made in this revision:-

Protection bypass format & checklist attached in annexure.

The revised LMI and COS OD documents are attached.

Submitted for approval of competent authority.

Submitted by:

तुषार श्रीवास्तव - वरिष्ठ प्रबंधक

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On:11/09/2023(06:22:46) **Ref:**SMTPP/OPERATION/2023-24/LMI REV/471660

Note No:2

LMI is reviewed and found in line with OD.

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On:12/09/2023(16:43:35) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:3

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On:15/09/2023(09:02:48) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:4

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On:15/09/2023(09:35:54) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:5

Review

Submitted by:

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On:15/09/2023(15:21:41) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:6

LMI Reviewed.

Forwarded for further review.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

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On:04/10/2023(18:57:26) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:7

Reviewed and submitted for approval

Submitted by:

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On:06/10/2023(09:05:35) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:8

Reviewed and submitted for further approval.

Submitted by:

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On:06/10/2023(16:42:52) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:9

1. Pl ensure that the registers kept in Control rooms (Main Plant and Offsite areas) are in line with the new annexures proposed in the revised LMI
2. Gist of this LMI especially changes if any, may be presented by the custodian (for the benefit of all sectional heads) at an early date in DPM and put up for further the approval

Submitted by:

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On:06/10/2023(17:28:05) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:10

Please give presentation at DPM.

Submitted by:

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On:07/10/2023(09:34:07) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:11

Presentation of LMI is given in DPM. LMI is put for further approval.

Submitted by:

तुषार श्रीवास्तव - वरिष्ठ प्रबंधक

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On:18/10/2023(20:52:48) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:12

It is a practice that ODs (Operation Directives) are to be followed in letter and spirit. Please confirm that the current LMI is atleast same as the OD (see copy in Annexures) in all respects. We can add to it, but cannot delete any thing from the OD. Operation incharge to confirm the same.

Please see the Revised LMI document in this light.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:27/10/2023(19:19:56) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:13

The proposed LMI is in line with OD. Necessary approval may please be acorded.

Submitted by:

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On:02/11/2023(15:31:46) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:14

LMI reviewed and found in order and in line with the corresponding OD

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

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On:04/11/2023(19:02:34) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:15

Initiator may please review as discussed. Also, please speak to Operation Incharge Sh. M.K. Jana, AGM(Opn.) regarding the same.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:06/11/2023(18:53:52) **Ref:**SMTPP/OPERATION/2023-24/LMI REV/471660

Note No:16

Suitable Modification done in Index of Revised LMI.

Put for approval.

Submitted by:

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On:14/11/2023(05:23:34) **Ref:**SMTPP/OPERATION/2023-24/LMI REV/471660

Note No:17

For review and necessary approval please.

Submitted by:

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On:16/11/2023(15:38:06) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:18

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On:21/11/2023(17:53:45) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:19

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On:23/11/2023(13:43:54) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:20

Review please

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

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On:24/11/2023(10:58:56) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:21

Submitted by:

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On:24/11/2023(11:48:02) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:22

Reviewed and found in order.

Put up for further approval.

Submitted by:

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On:03/12/2023(13:32:03) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:23

Reviewed and forwarded for further processing of the file.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

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On:03/12/2023(14:17:42) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:24

Reviewed and submitted for approval

Submitted by:

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On:05/12/2023(16:20:16) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:25

Kindly review and forward for further approval.

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On:05/12/2023(16:33:06) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:26

Reviewed & forwarded for approval

Submitted by:

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On:05/12/2023(16:48:31) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:27

May please be approved.

Submitted by:

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On:07/12/2023(12:03:26) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Note No:28

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On:08/12/2023(09:15:25) **Ref:**SMTTP/OPERATION/2023-24/LMI REV/471660

Local Management Instructions

Procedure to be followed for
Regulating/ Bypassing Permissive And Protections



Approved for implementation by HEAD OF PROJECT

Date 11.09.2023

Prepared by OPERATION DEPARTMENT

NATIONAL THERMAL POWER CORPORATION LTD.

Simhadri Thermal Power Station

LMI	Procedure to be followed for Regulating/ Bypassing Permissive And Protections		
LMI No	LMI/OD/OPS/SYST/005		
Issue No	1	Revision: 9	Dt: 11.09.2023

	Name	Designation	Signature
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Checked by:	UMA SHANKER VERMA	AGM(EEMG)	
Checked by	S K SUMAN	AGM(TMD)	
Checked by	OM PRAKASH	AGM(BMD)	
Checked by:	G.HARI NARAYANA RAO	AGM(C&I)	
Reviewed by:	T.R. RAJAGOPALAN	GM (OPN)	
Reviewed by:	MATHEW EIPE KOVOOR	GM(MAINT.)	
Reviewed by:	DHURJATI PRASAD PATRA	GM(O&M)	
Approved by:	SANJAY KUMAR SINHA	CGM(SIMHADRI)	

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1. INTRODUCTION

Permissives during start up and protection during operation of plant are provided to ensure the safety of plant, equipment and personnel. Many cases of catastrophic failures have been caused by either protection not acting as intended or being kept out of service. Whilst the former can be rectified by good engineering practices, the later can be avoided only by sound operation and maintenance practices. It is therefore of vital importance to ensure that all the permissive and protection of plant and equipment are always available and in service.

This LMI is issued to comply with operation directives OD/OPS/SYST/005. The aim of this LMI is to consider safety of plant, equipment and personnel by adopting sound operation and maintenance practices and this LMI offers directive on the implementation of this requirement.

2. SUPERSEDED DOCUMENTS & REVISION NOTES:

Superseded: LMI/OD/OPS/OPR/005 REV-08 dated Aug 2022.

Rev04 Oct'2011- Stage2 Units are added to scope

Rev 05 Dec'2017- Procedure bypass exception report format added as annexure-3

Rev06 Jan'2020: Revision notes introduced in paragraph 2.0.

Rev07 July'2020: Revision introduced in Clause No 5: Records of operator log entry included and Clause no:7 Monthly review by Head of Operation about all simulations done during the month and pending for removal at the end of the month, included.

Rev08 Aug'2022: LMI has been revised as per periodical review of 2 years period.

No modification made in this revision.

Rev09 Sept'2023: LMI has been revised as per Operation Directive "COS-ISO-00-OD/OPS/SYST/005 Rev. No.: 3 Date: AUGUST 2023".

Following modification is made: -Protection bypass format & checklist attached in annexure.

3. SCOPE:

The scope of this LMI covers the procedures to be followed for regulating the temporary bypassing of permissive / protections in the event of failure of any protection system components like Field device, relays, system I/O module, associated software, associated cabling etc. This procedure does not allow for continuous bypassing of any of the permissive/protections.

The scope covers: -

- a) Permissive and protection of boiler, turbine, generator, and switchyard.
- b) Permissive and protections of major auxiliary equipment (Refer Annexure-I).
- c) DC, HT and EHV systems and equipments.
- d) All other Auxiliary plants. (refer Annexure-II).

4.0 REFERENCE DOCUMENT:

This Location Management Instruction is based on:

- i) Operation Directive “COS-ISO-00-OD/OPS/SYST/005 Rev. No.: 3 Date: AUGUST 2023”
Procedure to be followed for Regulating/ Bypassing Permissives And Protections".

5.0 INSTRUCTIONS:

5.1 PERMISSIVE AND PROTECTION OF BOILER, TURBINE, GENERATOR AND SWITCHYARD

- 5.1.1 When all the channels provided in the logic for a particular protection/ permissive are not available, the associated unit should either be taken out of service or not returned to service until the protection is operative.
- 5.1.2 When the redundant channels provided in the logic for protection/ permissive are not available, the same may be bypassed for attending the defects after ensuring that the remaining channels are healthy and in service. This is to be done with the approval of Head of Operation and Maintenance, acting for HOS and defective channel should be restored in the shortest possible time.

5.2 PERMISSIVE AND PROTECTIONS OF MAJOR AUXILIARY EQUIPMENT.

- 5.2.1 When all the channels in the logic for protection/ permissive are not available in the equipment, the standby equipment should be taken into service and the concerned equipment taken out of service. Where no standby equipment is available bypassing of permissive/ protection may be carried out with the approval of Head of O&M acting for HOS. The protection/ permissive should be restored in the shortest possible time not exceeding a maximum of 8 hours.
- 5.2.2 When the redundant channel provided in the logic for protection/ permissive are not available, the same may be bypassed for attending to the defects after ensuring that the remaining channels are healthy and are in service and with the approval of Head of Operation acting for Head of O&M. The protection/ permissive should be restored in the shortest possible time not exceeding a maximum of 8 hours.

5.3 DC, HT AND EHV SYSTEMS AND EQUIPMENTS

- a) When all the channels provided in the logic for protection/ permissive are not available the system, equipment should be changed over with standby system/ equipment. However, if no standby equipment is available, bypassing of permissive/ protection may be carried out with the approval of Heads of O&M acting for HOS. The protection/ permissive should be restored at the earliest opportunity and in any case not exceeding a maximum of 8 hours.
- b) When the redundant channel provided in the logic for protection/ permissive are not available, the same may be bypassed for attending to the defects after ensuring that the remaining channels are healthy and are in service and with the approval of Head of Operation acting for Head of O&M. The protection/ permissive should be restored in the shortest possible time and in any case not exceeding a maximum of 8 hours.

5.4 PERMISSIVE AND PROTECTIONS OF ALL OTHER AUXILIARY PLANTS (INCLUDING OFF SITES AND CHP)

When all the channels in the logic for protection/ permissive are not available in the equipment in service, the standby equipment should be taken into service and the concerned equipment taken out of service. Where no standby equipment is available bypassing of permissive/ protection may be carried out with the approval of Head of Operation acting for Head of O&M. The protection/ permissive should be restored in the shortest possible time not exceeding a maximum of 8 hours.

When the redundant channel provided in one of the logic for protection/ permissive are not available, the same may be bypassed for attending to the defects, after ensuring that, the remaining channels are healthy and are in service. This operation may be completed with the approval of Shift Charge Engineer (Operation) acting for the in-charge of Operation. The protection/ permissive should be restored in the shortest possible time not exceeding a maximum of 8 hours.

6.0 NOTIFICATION OF BYPASSING OF PROTECTION

- a) Whenever the bypassing of protection/permissive is undertaken (as detailed in 5.1.2, 5.2.1 and 5.2.2.), information relating to the extent of the bypassing should be prominently displayed and must be entered in the Operations log book and the operator notified to ensure that all necessary measures are taken.
- b) If any Protection/Permissive is required to be bypassed due to any reason other than mentioned in the clause no 5.1,5.2 & 5.3, it must also be recorded as per clause no 7.0.

6.1 EXTENSION OF DURATION OF BYPASSING.

Whenever defective channel (s) could not be repaired and put into service for any reason within 8 hours as stipulated in the relevant paragraphs above, the duration of bypassing may be extended only after obtaining a recorded approval of HOP or by a reviewing authority who is one level higher than the authority to approve bypassing for 8 hours. However, the reviewing authority may accord his approval for continuing the bypassing only after taking into consideration the reasons for not being able to complete the repairs within 8 hours. This specific operating instructions and caution that has been given to the operating personnel etc.

7.0 RECORDS:

A record of protections / permissive bypassed with reason for bypassing and subsequent normalization shall be maintained at the following locations in Protections Bypass Register.

- a. Unit -I
- b. Unit-II
- c. Unit-III
- d. Unit-IV
- e. Switchyard control room
- f. Ash handling control room St-I
- g. Ash handling control room St-II
- h. CHP control room

The format for Protections Bypass & checklist is attached in annexure IV

8.0 REVIEW OF RECORDS

The records pertaining to the bypassing and restoration of protection shall be reviewed daily in the planning meeting by head of O&M. All simulations done during the month or, pending for removal at the end of the month, shall be reviewed and signed off by the head of Operation.

Exceptions and simulations report on the status of unit protections and switchyard shall be put up by Head of O&M to Head of station for his perusal and the same shall be monitored in ORT. The format for exception report is attached at Annexure-1.

The records pertaining to the bypassing and restoration of protection shall be reviewed daily by the heads of operation and maintenance & at least once in a month by Head of Station. Observations shall be specifically recorded if undue delays are noticed in restoration, or if, protection devices are being bypassed repeatedly.

Exception report on the status of unit protection and switch yard shall be put up by Head of O&M to the Head of Station for his perusal. The format for Exception Report is attached (Refer Annexure-III)

9.0 RESPONSIBILITY

Head of Operation is responsible for implementation of this LMI.

10.0 REVIEW:

This LMI is to be reviewed every Two years by the Head of Operation

ANNEXURE-I

Auxiliary equipment main plant	
Sr.No.	Equipment
1	Induced draft Fan
2	Forced Draft Fan
3	Primary Air Fan
4	Air preheater
5	Coal mill
6	Boiler circulating water pump
7	Boiler feed pumps
8	Condensate extraction pumps
9	DM CW pumps
10	ARCW pumps
11	HP/LP feed water heaters
12	Aux. oil pump
13	Jacking oil pump
14	Vacuum pump
15	Seal oil pumps
16	Primary water pump
17	Circulating water pump
18	Instrument air compressor
19	Plant air compressor
20	Sea water pump
21	Electro chlorinator.
22	Sweet water pumps
23	Ash handling transport air compressor
24	Ash disposal pump
25	Bottom ash HP and LP water pumps.
26	Fly ash HP water pumps.
27	Effluent disposal pumps
28	Fire water pumps.
29	Diesel generator sets.

ANNEXURE - II

Auxiliary equipment main plant	
Sr.No.	Equipment
1	Fuel oil handling system
2	Air-conditioning system
3	Ash water recirculation pumps.
4	Mill rejects handling system
5	Dry ash handling and silo equipment.
6	Service water system
7	Potable water system
8	Fire fighting system.
9	Oil purification systems.
10	Raw water system
11	Pre treatment plant
12	DM plant
13	Effluent treatment plant
14	Ash handling system including seal water and LP water pumps.
15	Coal Handling Plant

Annexure-III

Exception report on the status of protections:

BOILER	Unit No.	Brief Reason with duration
Protections leading to MFT		
TURBINE		
Protections leading to Turbine trip		
GRP		
Protections leading to Generator Class A/Class-B trip		
SWITCH YARD	Description	
Line Protection		
Bus Bar protection		
Protection related to Offsite and CHP	Equipment	

Head of O&M

Head of Station

Protection bypass Format

Bypassing of Protection/Permissive		
1	Name of Protection/Permissive to be bypassed	
2	Equipment	
3	Reason for Bypassing	
5	Requested by	
6	Name & signature of the Person authorizing Bypass of protection /Permissive	
7	Name & signature of Person bypassing the protection/permissive	
8	Date & time of Bypassing	
9	Expected time required for Restoration	
Restoration of Protection/Permissive		
1	Date & Time of Restoration	
2	Name & signature of person restoring the Protection/Permissive	
3	Acknowledgement of SCE	
4	Reviewed by	

Checklist for Protection Bypass

S.No	Description	Status
1.	If any unit protection is to be bypassed, ensure corresponding back up protection is in service.	
2	If any unit protection is to be bypassed, standard instruction (SOP) is in place if trip condition persists corresponding to the protection.	
3	Healthiness of alternate channels is checked before bypassing any protection.	
4	Post bypassing, confirm with C&I that intended protection is only bypassed.	
5	Post normalization of any protection, confirm with C&I that protection is in service.	
6	If P&I checks are done for Unit protection or on any equipment then post checking all protection values are restored to recommended values.	
7	If any permissive is bypassed the same to be recorded in Protection bypass register	
8	During normalization of any protection, check that trip condition is not prevailing.	
9	When unit is being brought back after long outage or AOH, protection status to be crosschecked by at least two levels Operation general shift & SCEs.	
10	Flame failure protection should never be bypassed under any circumstances if boiler is live.	